

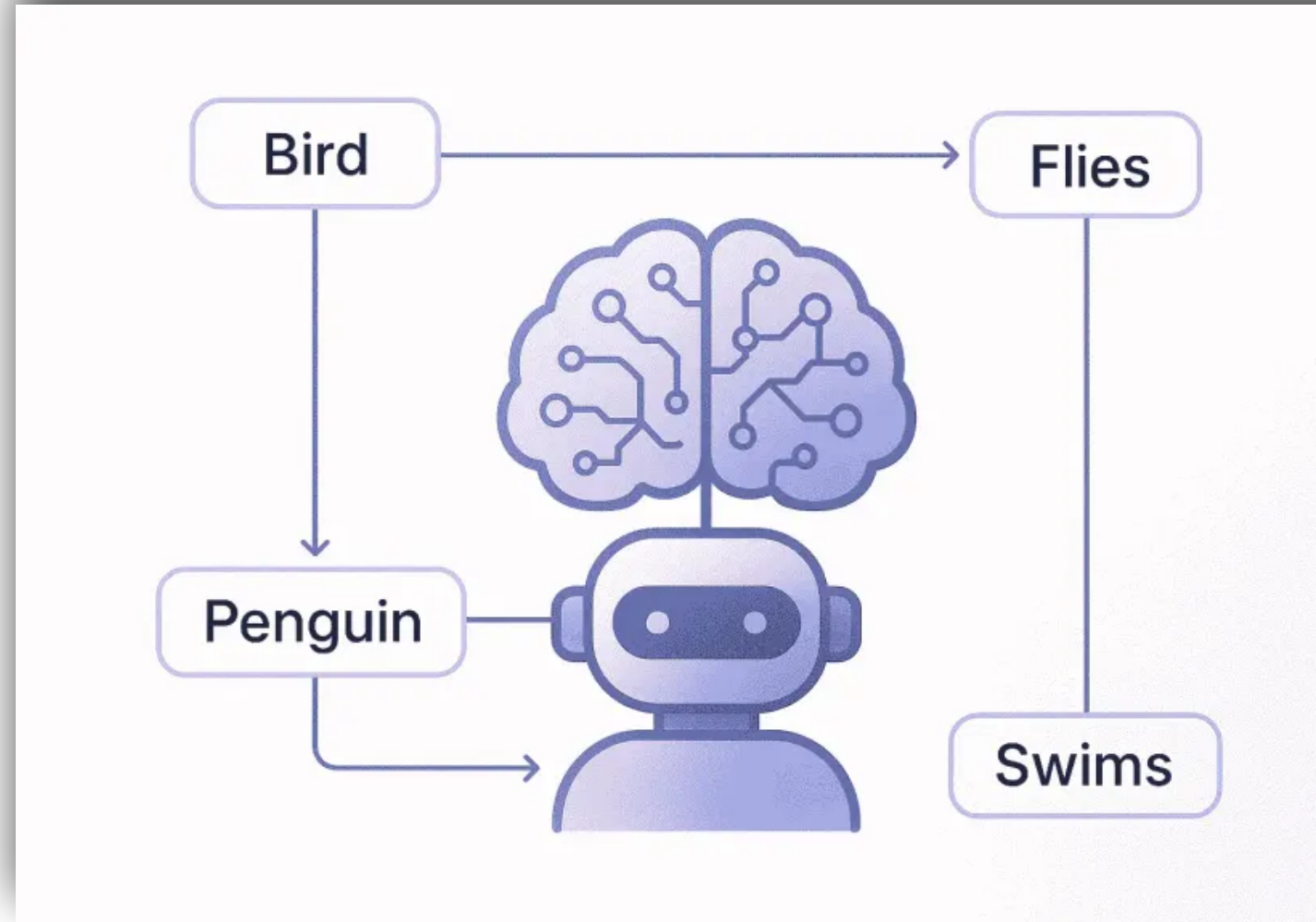
WEEK 4 – LECTURE 1

Artificial Intelligence Foundations

Comparing Search Algorithms & Knowledge Representation

Semantic Networks • Frames

CSE276 | Unit II



Quick Recap

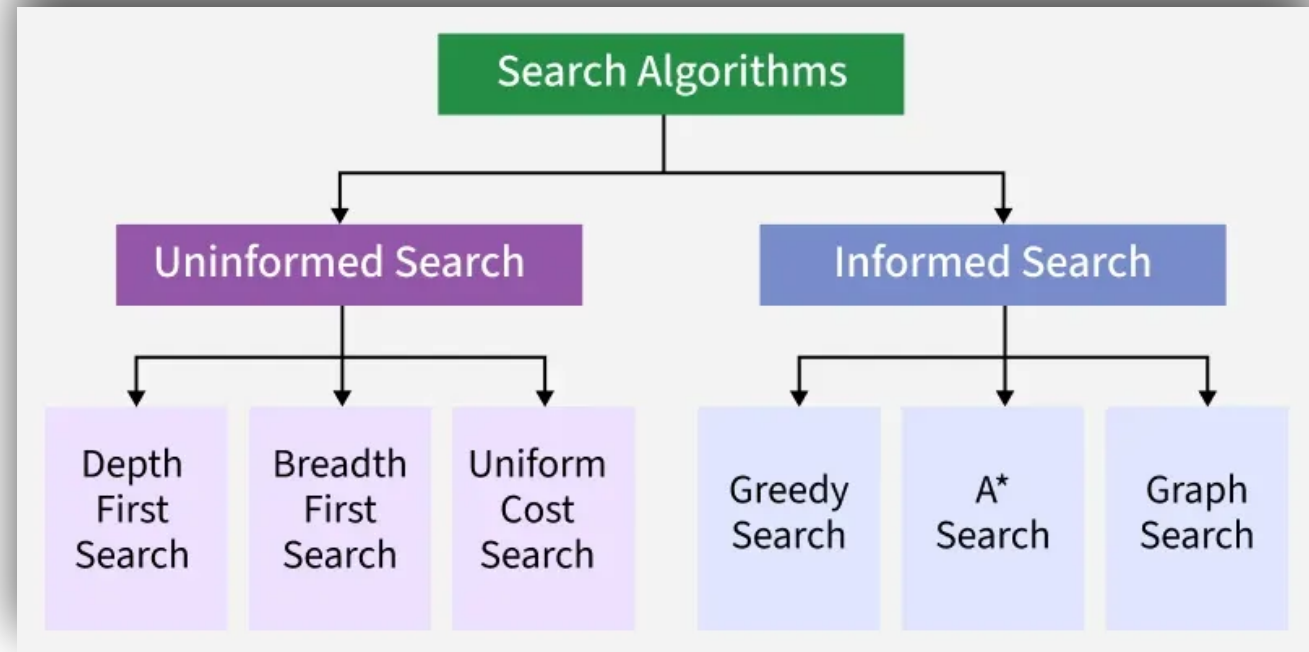


Which search algorithm should we use for a given problem?

Why Compare Search Algorithms?

Algorithms solve problems differently.

- 🔍 Some are fast but ignore optimal paths.
- 🏠 Some find the best path but use too much memory.
- 🧠 Some require hints (heuristics) to work well.



Are all search algorithms equally good?

Comparing Search Algorithms

Algorithm	Strategy	Heuristics?	Shortest Path?	Memory
BFS	Level-by-level	✗	✓ (Unweighted)	High
DFS	Deep dive	✗	✗	Low
Best-First	Closest estimate	✓	✗	Medium
Hill Climbing	Immediate neighbor	✓	✗ (Local Max)	Very Low
A*	Cost + Estimate	✓	✓	High

Which Algorithm Should We Choose?



**GPS
Navigation**

A*



**File
System
Search**

DFS



**Social
Network
Friends
(1st
degree)**

BFS



**Optimizing
an
Equation
(No
memory)**

Interactive Activity

If you are running an AI on a small embedded device with very limited memory, which search algorithm would you choose? Why?

BFS

A*

Search Algorithms



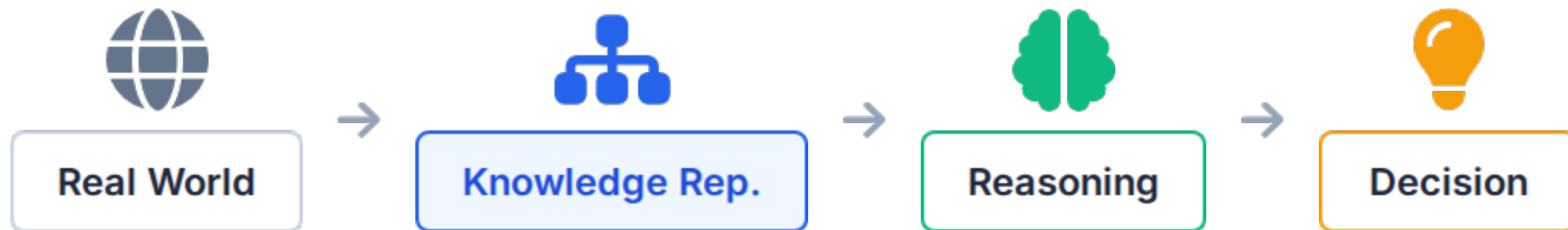
Find Information



But where does AI store the information it finds?

What is Knowledge Representation?

AI needs a structured way to store facts to reason about the world.



Why Represent Knowledge Formally?

Without Structure



Random notes scattered everywhere.



Confusion / Hard to Search

With Structure



Organized notebook / encyclopedia.



Easy Retrieval / Logic

AI cannot reason from unstructured information alone.

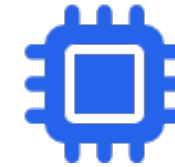
Real-World Analogy

Human Brain

Organizes memories



Recalls information to act



AI System

Stores structured knowledge



Uses it to answer questions

Types of Knowledge Representation

Knowledge Representation



Semantic Networks

Graphs & Links



Frames

Slots & Attributes



Rules

IF-THEN Logic

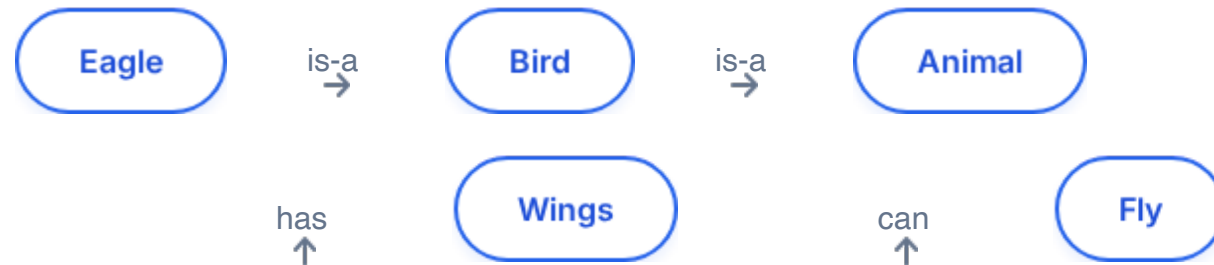


Formal Logic

Mathematical Logic

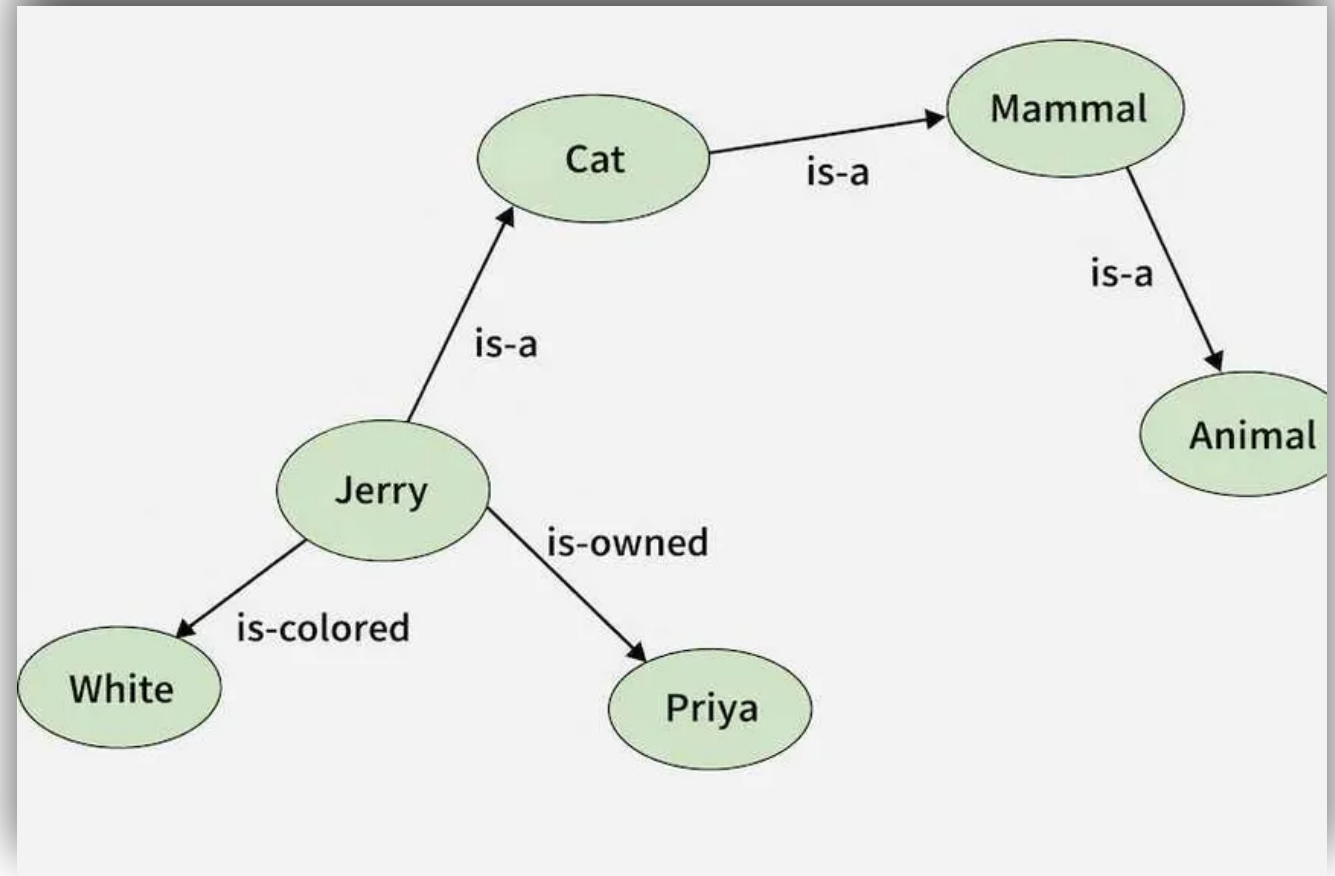
Semantic Networks

Represents knowledge as a graph of interconnected nodes and labeled edges.



Semantic Network Example

Jerry is a Cat.
Jerry is a Mammal.
Jerry is an Animal.



Connected knowledge allows AI to infer answers.

Frames

- 👤 Stores all info about one object together.
- 📋 Uses a structure like a form or template.
- 🏷️ Composed of "Slots" (Attributes) and "Values".

Frame: Student

Name:Alice Smith

Roll No:CS101

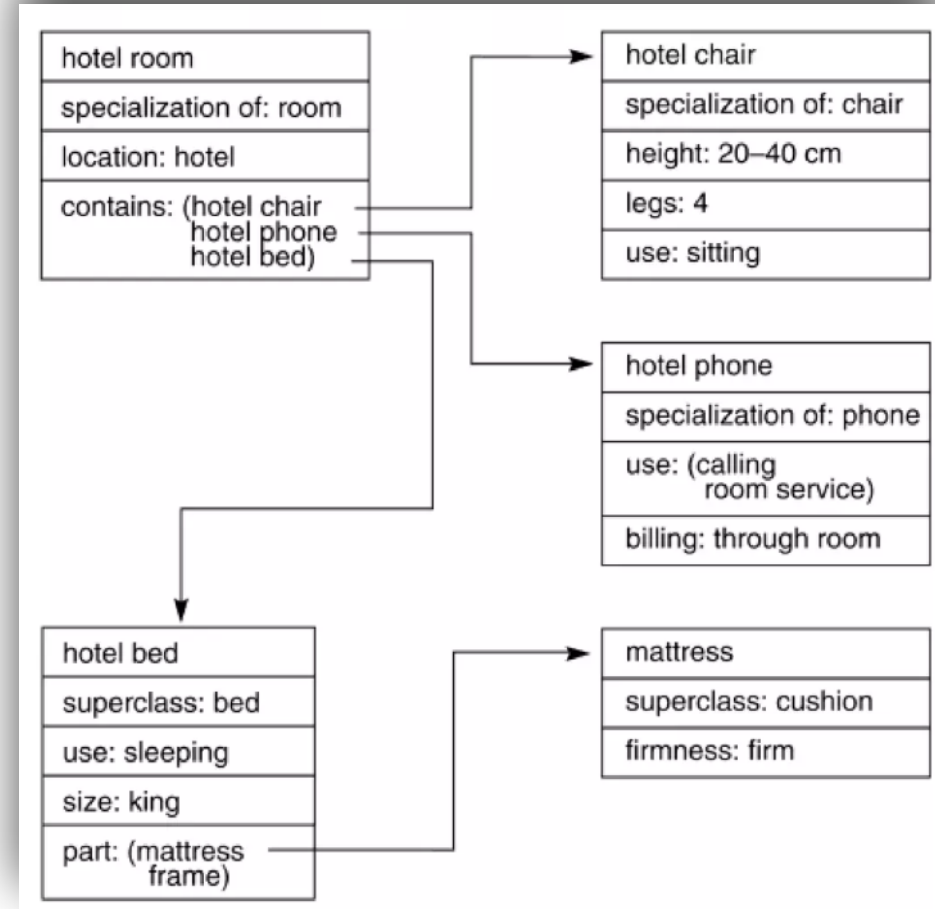
Dept:Computer Science

Semester:1st

CGPA:3.8

Frame Example

Frame: Car
Manufacturer: Toyota
Fuel Type: Hybrid
Color: Silver
Speed: 180 km/h
Owner: Bob



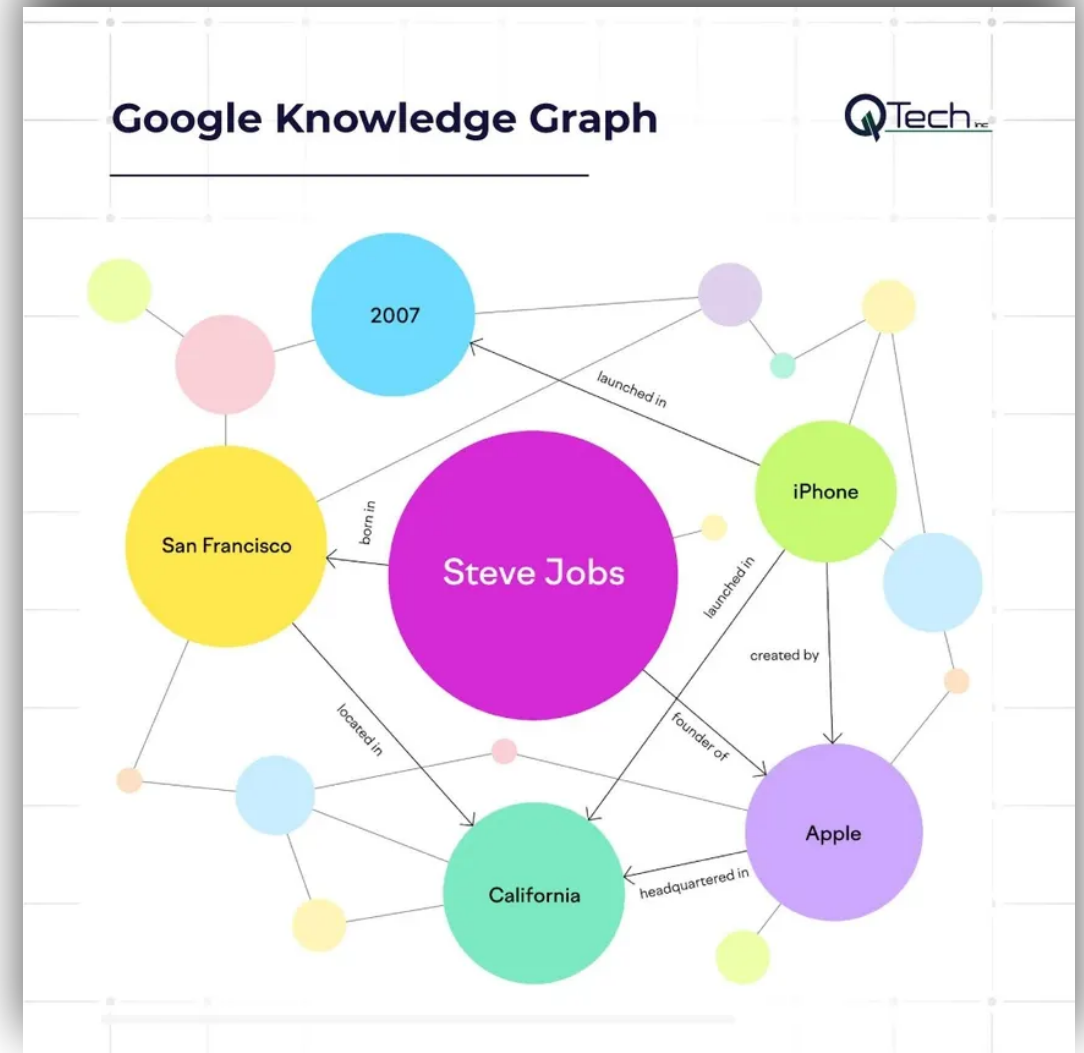
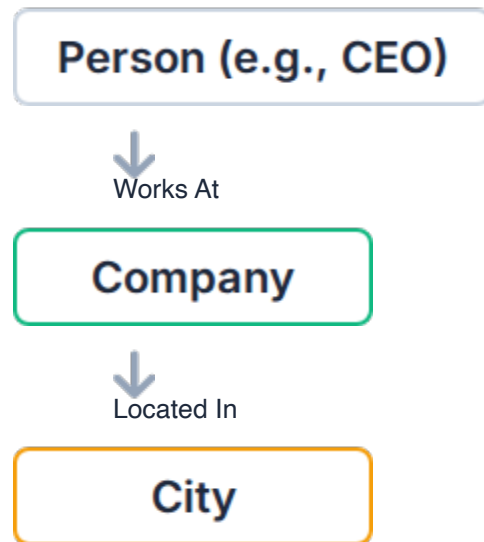
Similar to a database record or object in programming.

Semantic Networks vs Frames

Feature	 Semantic Network	 Frames
Style	Graphs & Links	Templates & Slots
Focus	Relationships between concepts	Attributes of a specific object
Structure	Flexible, distributed	Rigid, centralized
Analogy	Mind Map / Family Tree	Form / ID Card / DB Record

Case Study: Google Knowledge Graph

When you search for a famous person, Google doesn't just read text; it uses a massive Semantic Network.





Board Activity

Instructor Draws:

-  1. A Semantic Network for a University
-  2. Converts the same information into Frames

Common Misconceptions



"Semantic networks are just databases."

They represent concepts and relationships, not just tables of data.



"Frames are identical to spreadsheets."

Frames can inherit properties from other frames (like Object-Oriented Programming).



"Search and Knowledge Representation are the same thing."

Search finds the path; KR stores the facts needed to understand the path.

Key Takeaways



Different search algorithms suit different problems.



AI must formally represent knowledge to reason.



Semantic Networks focus on connections.



Frames organize attributes of specific objects.

Quick Quiz

Which knowledge representation method organizes information using "Slots" and "Values"?

- ☐ A) Semantic Networks
- ☐ B) Frames
- ☐ C) Breadth-First Search
- ☐ D) Uninformed Search

Quick Quiz

Which knowledge representation method organizes information using "Slots" and "Values"?

- ☐ A) Semantic Networks
- ☒ **B) Frames**
- ☐ C) Breadth-First Search
- ☐ D) Uninformed Search

Preview: Next Lecture

Production Systems & Expert Systems

- ☰ Rules and IF-THEN Reasoning
- 🗄 Knowledge Base
- ⚙ Inference Engine

